Jad Haddad

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EDUCATION

Texas A&M University — **College Station, Texas**
Bachelor of Science in Electronic Systems Engineering | 3.3 GPA — *Aug. 2023 - Dec. 2027*

Relevant Coursework: Power Systems & Circuits, Analog & Digital Electronics, Microcontroller Architecture, Embedded Systems Design

Certifications: Lean Six Sigma Yellow Belt (Texas A&M); Cloud Computing: Core Concepts (NASBA)

EXPERIENCE

Oncor Electric Delivery, Glen Rose Transmission — **Glen Rose, Texas**
Protection & Control Engineering Intern — *May 2026 – Aug. 2026*

- Programmed, configured, and field-tested protective relays across electromechanical, solid-state, and microprocessor-based generations, supporting protection and control across 345kV and 138kV transmission substations serving the Comanche Peak Nuclear Power Plant.
- Performed secondary current-injection and insulation-resistance testing on Doble, Arbiter, and Omicron test sets, validating pickup thresholds, trip paths, and scheme operation against approved settings and documenting results for engineering review.
- Delivered a technical capstone on capacitor-bank fundamentals of theory and application, presenting to 25+ directors and senior engineers and building 4 interactive HTML/SVG simulations adopted as part of a new, company-wide standard training reference on capacitor banks.

ZKK Consulting LLC — **Dallas-Fort Worth Metroplex, Texas**
Co-Founder, Technical & Patent Research Lead — *Aug. 2026 – Present*

- Co-founded an engineering research firm and built its technical practice across electronics, embedded systems, connected products, component and supply-chain landscapes, patent landscapes, and certification pathways, generating ~$15,000 within the first month.
- Lead patent and prior-art research across medical devices, electronics, and engineering-led hardware, mapping up to 15 patents in a single landscape across technical features, ownership, and status to clarify market-entry decisions before capital is committed.
- Build bottom-up TAM, SAM, and SOM models and competitive teardowns, translating component cost, supplier availability, and FCC & FDA certification pathways into a defensible market-entry cost and risk picture.

ENGINEERING PROJECTS

Vyrex, Wrist-Worn Fall-Detection Wearable — **College Station, Texas**
Founder, Hardware Lead — *Feb. 2026 – May 2026*

- Designed a compact (30x25mm) 4-layer mixed-signal PCB in KiCad, integrating an ESP32-C3 Mini BLE microcontroller, LSM6DSV16XTR 6-axis IMU, Li-Po charge management, USB-C regulation, and MOSFET-driven haptics with flyback protection.
- Evaluated 3 MCUs across availability, power draw, and pin compatibility, selecting a pre-certified MCU module that eliminated FCC 47 CFR Part 15C intentional-radiator testing, decreasing testing fees by ~$17,000.
- Implemented on-device fall detection using the LSM6DSV16XTR Machine Learning Core, reducing data offloads and deep-sleep interrupts while enabling gait-based parameter retraining through a companion BLE app.
- Developed a full BOM across JLCPCB's supply chain and validated the architecture on an ESP32 development kit before the final board spin.

Smart Table Solutions, Height-Adjustable Desk Control System — **Plano, Texas**
Founder, Self-Employed Electronics Engineer — *May 2025 – Jul. 2025*

- Designed and sold an STM32-based motor-controlled table, generating ~$2,400 across 8 customer units at a 100% on-time fulfillment rate.
- Engineered a 12V H-bridge motor driver with overcurrent protection, low-pass gate filtering, and closed-loop PWM speed control, validated the design in LTspice before board bring-up.
- Implemented a modular PCB in KiCad with spare I/O for custom add-ons and control firmware and designed the enclosure in Solid Edge.

SoilSense, Automated Irrigation Controller — **McKinney, Texas**
Personal Project — *Apr. 2025 – Jul. 2025*

- Built an Arduino Uno R3 irrigation controller in C++ with capacitive soil-moisture sensing, two-channel N-MOSFET pump switching, ultrasonic reservoir-level monitoring, and LED dry-run alerting, running 9+ months without intervention or failure.

LEADERSHIP & INVOLVEMENT

Institute of Electrical and Electronics Engineers (IEEE) — **College Station, Texas**
Student Member — *Dec. 2025 – Present*

- Apply chapter workshop training in Keil µVision and KiCad to personal hardware projects for firmware bring-up and PCB layout.

Professional Association of Industrial Distribution (PAID) — **College Station, Texas**
Chapter Member — *Dec. 2025 – Present*

- Apply supplier network and supply-chain economics training from chapter workshops to component sourcing and BOM cost work.

Freshmen Exemplifying Aggie Spirit Together (FEAST) — **College Station, Texas**
Mentor & Former Special Events Planning Committee Member — *Aug. 2023 – Present*

- Mentor 15 undergraduate peers through their academic, organizational, and personal transition into Texas A&M; previously managed an ~$8,000 semester budget and event logistics for 96 peers on the Special Events Planning Committee.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, ARM Assembly, Bash, MATLAB, Verilog, JavaScript, HTML/CSS, Tailwind

Hardware: Mixed-signal PCB design, power management & Li-Po charging, IMU & sensor integration, BOM development/sourcing

Embedded: STM32, ESP32, BLE, UART, I2C, SPI, GPIO, PWM, ADC/DAC, ESP-IDF, low-power/deep-sleep design, Edge Machine Learning

Tools: AcSELerator QuickSet, Powerbase, ProjectWise, Multisim, Excel, Wireshark, STM32 Cube Programmer, Doble RTS, Linux, Arduino IDE